

22.10.2024

EXPERIMENT-9

Fraud Detection (Classification)

```
1 import pandas as pd
2 from imblearn.over_sampling import SMOTE
3 from sklearn.preprocessing import StandardScaler
4 from sklearn.model_selection import train_test_split
5 from sklearn.svm import SVC
6 from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, classification_report
```

```
1 # Load dataset
2 data = pd.read_csv('creditcard.csv')
```

```
1 # Split features (X) and target (y)
2 X = data.drop('Class', axis=1)
3 y = data['Class']
4
5 # Apply SMOTE to handle class imbalance
6 sm = SMOTE(random_state=42)
7 X_res, y_res = sm.fit_resample(X, y)
8
9 # Standardize the feature columns
10 scaler = StandardScaler()
11 X_scaled = scaler.fit_transform(X_res)
```

```
1 # Split the data into training and test sets
2 X_train, X_test, y_train, y_test = train_test_split(X_scaled, y_res, test_size=0.2, random_state=42)
```

```
1 # Initialize the SVM model with RBF kernel and balanced class weight
2 svm_model = SVC(kernel='rbf', class_weight='balanced', random_state=42)
3
4 # Train the SVM model
5 svm_model.fit(X_train, y_train)
6
7 # Predict on the test set
8 y_pred = svm_model.predict(X_test)
```

```

1 # Calculate evaluation metrics
2 accuracy = accuracy_score(y_test, y_pred)
3 precision = precision_score(y_test, y_pred)
4 recall = recall_score(y_test, y_pred)
5 f1 = f1_score(y_test, y_pred)
6
7 # Print the evaluation results
8 print(f"Accuracy: {accuracy}")
9 print(f"Precision: {precision}")
10 print(f"Recall: {recall}")
11 print(f"F1 Score: {f1}")
12 print("Classification Report:\n", classification_report(y_test, y_pred))

```

Accuracy: 0.9959200182895732

Precision: 0.9977627453052884

Recall: 0.9940852288682954

F1 Score: 0.9959205922174746

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	1.00	56750
1	1.00	0.99	1.00	56976
accuracy			1.00	113726
macro avg	1.00	1.00	1.00	113726
weighted avg	1.00	1.00	1.00	113726